

CODED BEAM TRAINING

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Abstract

XL-MIMO is a key technology for future 6G systems requiring efficient beam training for CSI acquisition.

Conventional codebook-based methods perform poorly for low SNR users, causing rate degradation.

A coded beam training approach is proposed by integrating channel coding concepts.

A duality between hierarchical beam training and channel coding is established.

A general framework is developed with implementations using Hamming and convolutional codes. Beam encoding and decoding are optimized for improved performance.

Results show enhanced accuracy, extended coverage, and low training overhead.

Introduction

XL-MIMO is a key technology for 6G communication systems due to its ability to provide high data rates and spectral efficiency. However, accurate beam alignment is essential for effective communication. Beam training is used to identify the best beam direction between the base station (BS) and user equipment (UE) [1].

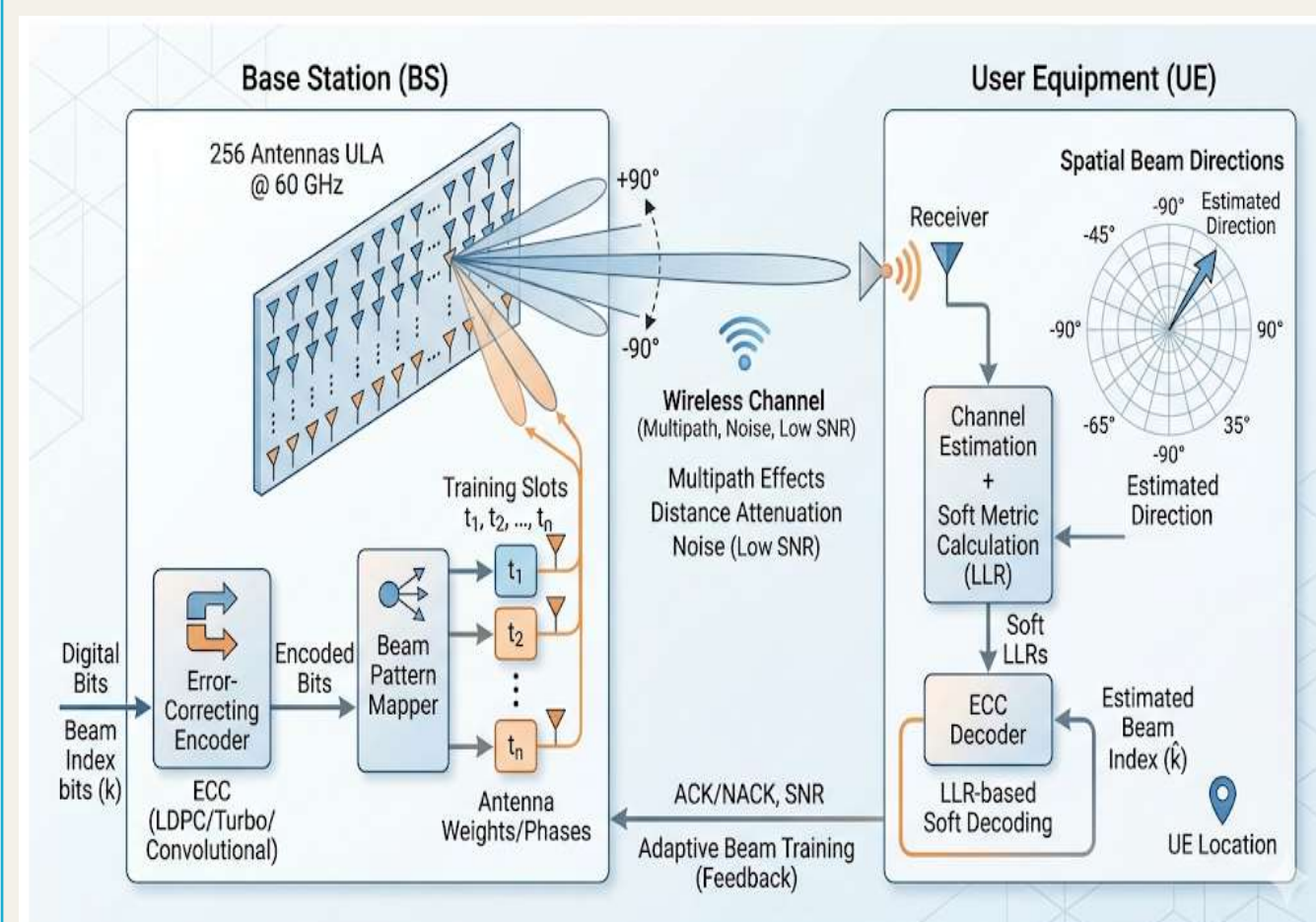
Traditional beam training methods include exhaustive search and hierarchical search. Exhaustive beam training provides high accuracy but requires very high training time, while hierarchical beam training reduces overhead but suffers from error propagation, especially in low SNR conditions [2].

This problem becomes severe for cell-edge users due to weak signals and noise. According to Shannon's theory, the achievable transmission rate is limited by channel capacity, which depends on SNR, highlighting the difficulty of reliable communication in low-SNR scenarios [5].

Our work focuses on improving beam training performance using coded beam training. Unlike the base approach, we extend it by evaluating advanced error-correcting codes are widely used for reliable communication in noisy channels [4].

We also incorporate adaptive beam training and soft-decision decoding, which use likelihood-based metrics instead of hard decisions. This significantly improves robustness against noise. The proposed system achieves higher success rate, improved sum rate, and extended coverage compared to existing methods [3].

System Model



The received signal at the user is given by::

$$y = h^H w x + n$$

The optimal beamforming vector is selected to maximize the received signal power:

$$w^* = \arg \max_{w \in \mathcal{W}} |h^H w|^2$$

The performance of the system is evaluated using SNR, defined as:

$$\text{SNR} = \frac{|h^H w|^2 P}{\sigma^2}$$

To improve reliability, the beam index is first represented as a binary sequence $\mathbf{b}$, which is encoded using a channel coding function:

$$c = \mathcal{E}(\mathbf{b})$$

Due to channel impairments, the received codeword is:

$$r = c + e$$

where e represents the error introduced by noise.

At the receiver, decoding is performed to recover the original beam index:

$$\hat{\mathbf{b}} = \mathcal{D}(r)$$

The use of channel coding reduces the probability of incorrect beam detection:

$$p_e^{\text{coded}} < p_e^{\text{uncoded}}$$

The achievable data rate of the system is given by:

$$R = \log_2 \left(1 + \frac{|h^H w|^2 P}{\sigma^2} \right)$$

Results & Discussion

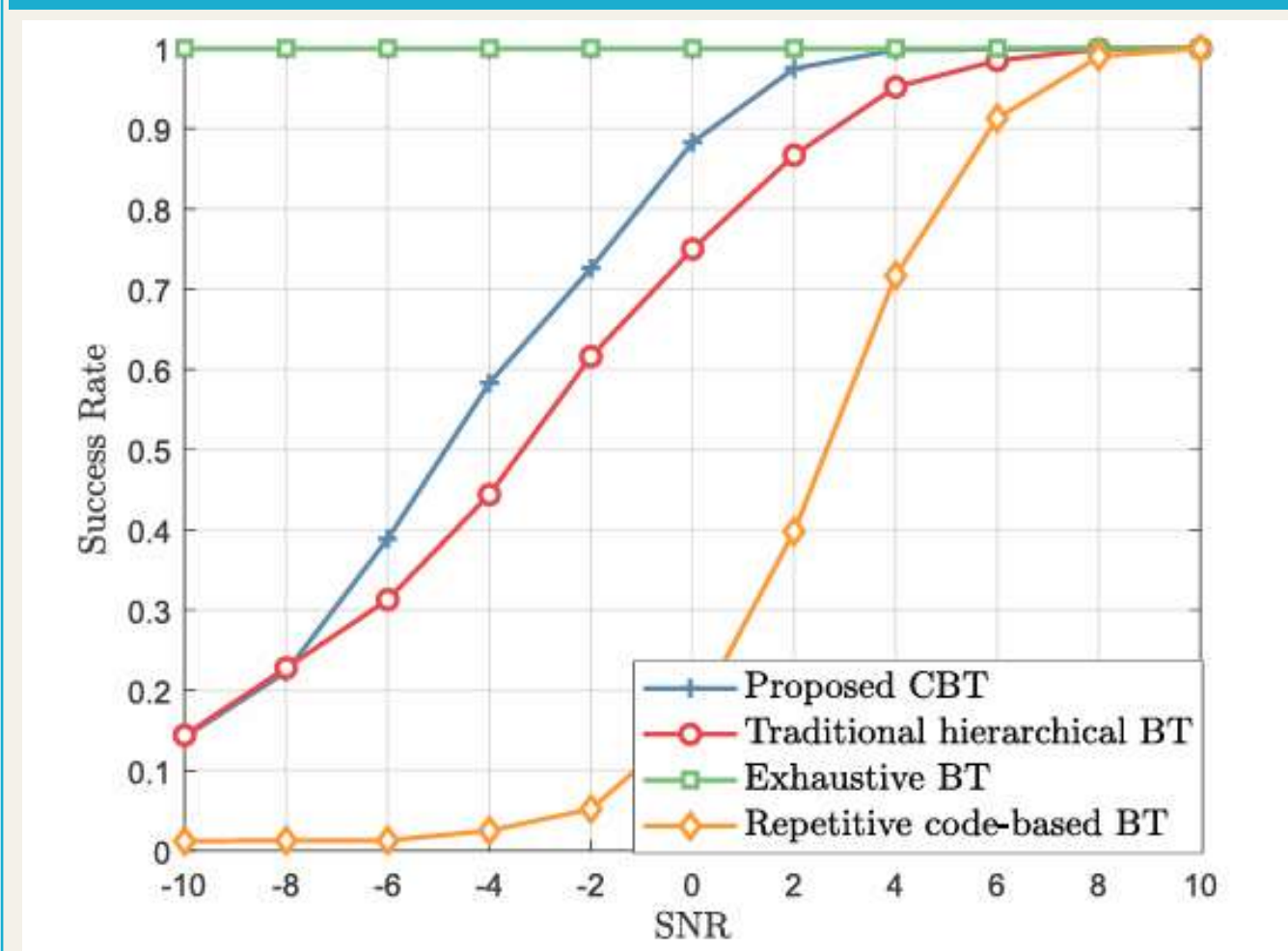


Fig. 1. Comparisons of the success rate for different beam training methods.

Success Rate vs SNR

- Proposed CBT shows significantly higher success rate than traditional methods, especially at low SNR.
- Exhaustive Beam Training achieves near-perfect success rate across all SNR values but at high complexity.

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- Traditional Hierarchical BT improves gradually but performs worse than CBT at all SNR levels.
- Repetitive Code-based BT performs poorly at low SNR and only improves at high SNR.

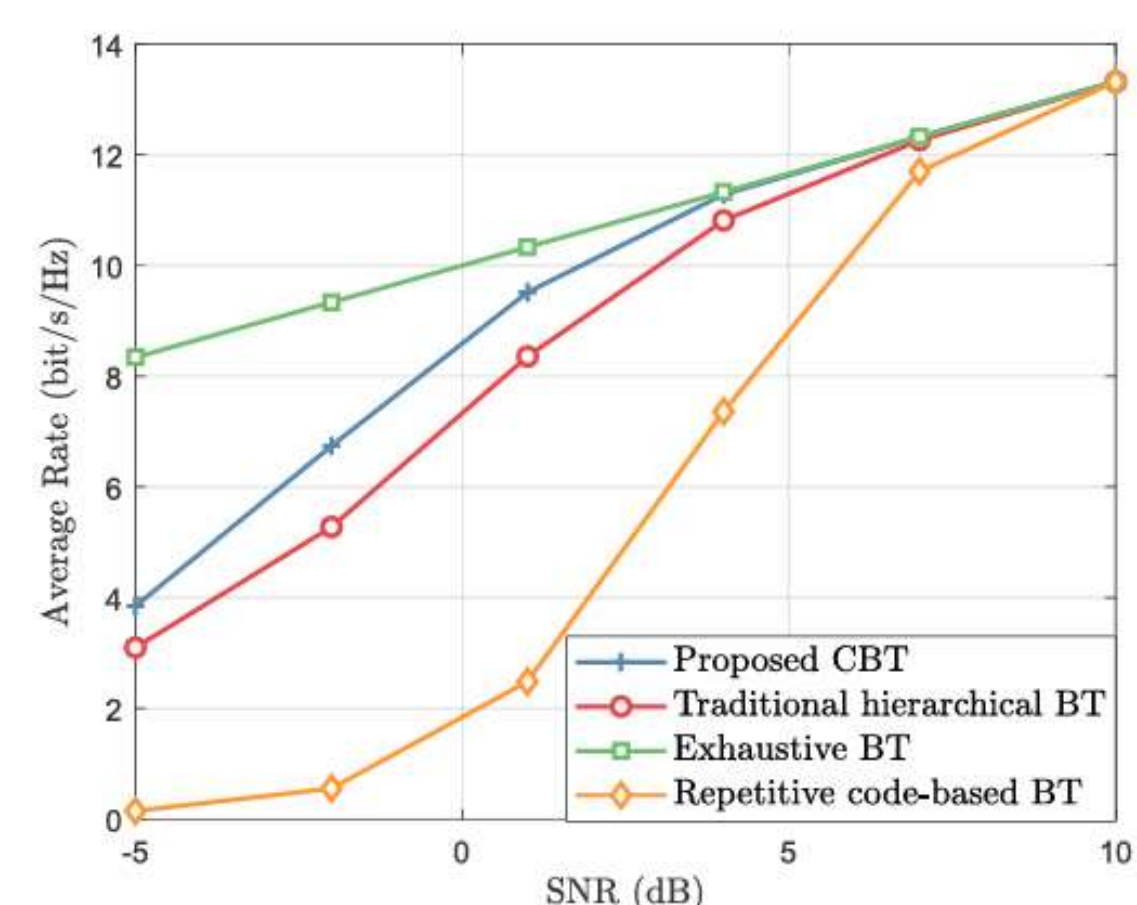


Fig. 2. Comparisons of the average rate for different beam training methods in mixed LoS and NLoS channel

Average Rate vs SNR

- Proposed CBT achieves higher data rates than hierarchical methods, especially in low to moderate SNR.
- Exhaustive BT provides the highest rate but is impractical due to large overhead.
- Traditional Hierarchical BT shows moderate performance but lags behind CBT.
- Repetitive Code-based BT has very low rates at low SNR and improves only at higher SNR.

Conclusion & Future work

Conclusion

- Proposed coded beam training (CBT) improves beam alignment accuracy and reliability in XL-MIMO systems, especially under low SNR conditions.
- Achieves better performance than hierarchical methods with low overhead and extended coverage.

Future Work

- Extend the approach to multi-user and dynamic channel environments.
- Explore advanced techniques like AI-based beam prediction and stronger channel codes.

References

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